Toward a Canonical Residual–Force Scaling Law for Coherent Nonequilibrium Electromagnetic Systems

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Abstract

A phenomenological residual-force scaling framework for coherent nonequilibrium electromagnetic systems is proposed. The framework emphasizes asymmetry, resonance coherence, artifact discrimination, stress-energy accounting, and experimentally testable scaling relations. A canonical EVRT residual-force equation is introduced together with a formalized experimental metrology framework intended to constrain or experimentally bound an effective residual-response coefficient. No experimentally verified residual-force effect is claimed in this work.

1 Introduction

Equation (1) should be interpreted as a constrained phenomenological scaling relation rather than a first-principles microscopic derivation.

The central unresolved question considered in this work is:

> Can coherent nonequilibrium electromagnetic architectures produce residual effects that survive complete artifact rejection?

This work does not claim established anomalous propulsion. Robust null results remain scientifically valuable because they constrain the admissible parameter space of nonequilibrium residual-response models.

Instead, the present work develops a constrained phenomenological framework intended for precision metrology, falsifiable testing, and systematic artifact discrimination.

2 Canonical EVRT Residual–Force Equation

$$F_{\mathrm{res}} = \chi_{\mathrm{eff}}\, u_{\mathrm{EM}}\, V\, \mathcal{A}\, \mathcal{Q}\, \mathcal{G} \tag{1}$$

The coefficients $\mathcal{A}$, $\mathcal{Q}$, and $\mathcal{G}$ are dimensionless phenomenological scaling factors intended to parameterize asymmetry, coherence, and boundary-condition coupling respectively.

The proposed scaling relation is intended only as an experimentally parameterized effective model and should not be interpreted as a derivation from established microscopic field theory. with

$$u_{\mathrm{EM}} = \frac{1}{2}\left(\epsilon_0 E^2 + \frac{B^2}{\mu_0}\right). \tag{2}$$

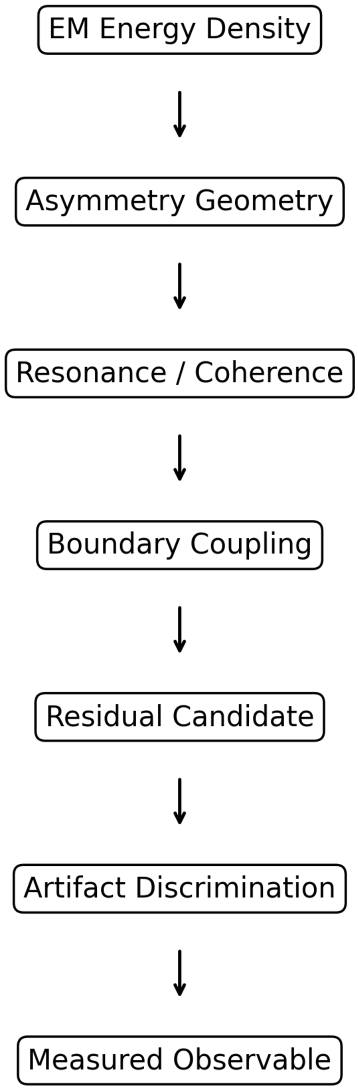

Figure 1: Conceptual EVRT framework progression from electromagnetic energy density through asymmetry, coherence, boundary coupling, and artifact discrimination toward measurable observables.

3 Phenomenological Derivation Pathway

The proposed equation is motivated through the scaling pathway:

$$u_{\mathrm{EM}} \to U_{\mathrm{field}} = u_{\mathrm{EM}} V \to \text{stress asymmetry} \to F_{\mathrm{res}}.$$

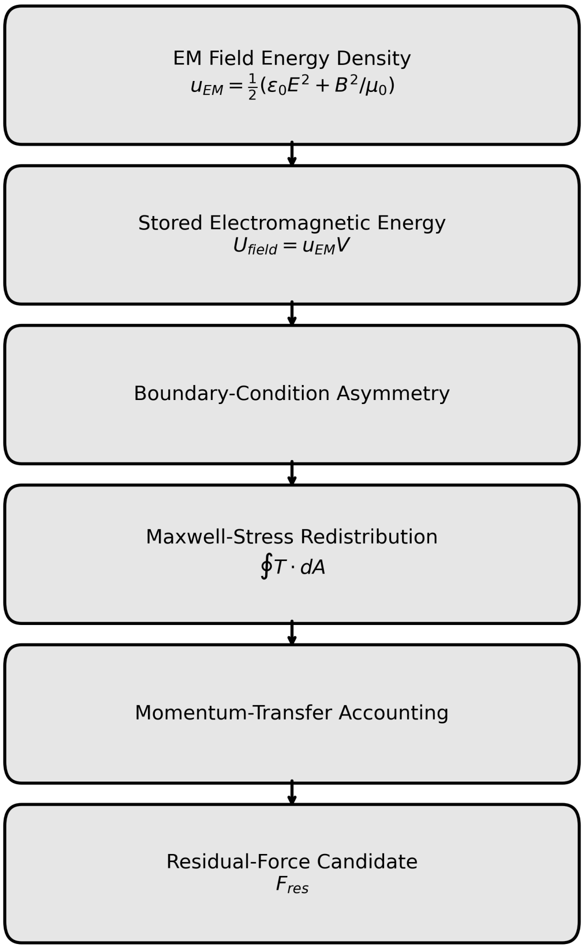

Figure 2: Illustrative phenomenological derivation pathway connecting electromagnetic energy density, boundary-condition asymmetry, Maxwell-stress redistribution, and momentum-transfer accounting to candidate residual-force observables.

The framework explores whether strongly asymmetric nonequilibrium field configurations may produce an effective residual response parameterizable through χ_{eff} after conventional force channels are independently constrained.

4 Stress–Energy and Momentum Accounting

A useful bridge to established electrodynamics is the Maxwell stress tensor formulation,

$$\mathbf{F} = \oint \mathbf{T} \cdot d\mathbf{A},$$

together with stress-energy conservation,

$$\partial_\mu T^{\mu\nu} = 0.$$

In a closed symmetric system, internal stresses should cancel unless momentum is transferred through radiation, plasma, environmental coupling, mechanical supports, or measurement artifacts.

Any experimentally inferred residual must therefore be interpreted relative to the cumulative uncertainty associated with all known momentum-transfer pathways.

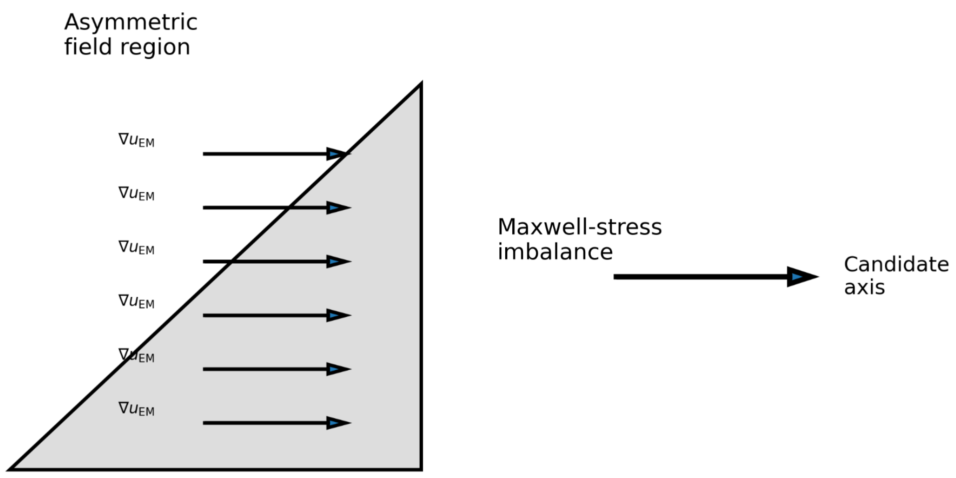

Figure 3: Conceptual illustration of asymmetric electromagnetic stress distributions and candidate stress-imbalance axes within a nonequilibrium field geometry.

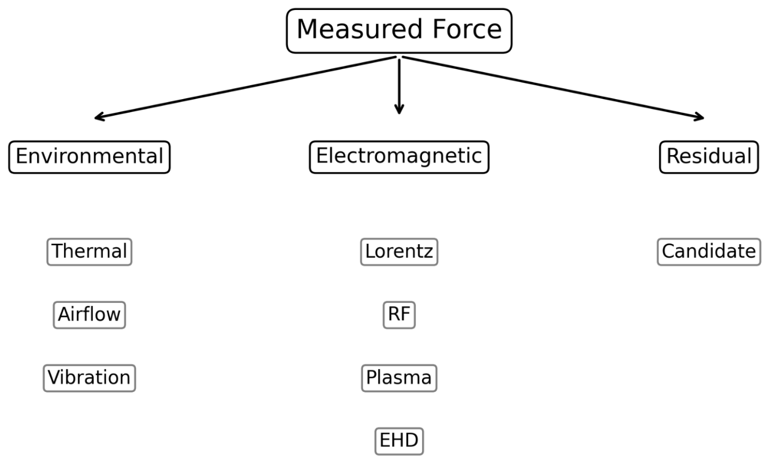

Figure 4: Illustrative decomposition of candidate force signatures into environmental, electromagnetic, and residual diagnostic channels.

5 Operational Parameter Definitions

Table 1: Operational interpretation of EVRT parameters.

Parameter	Operational definition
χ_{eff}	inferred coefficient or experimental upper bound
u_{EM}	electromagnetic energy density
V	active field volume
$\mathcal{A}$	asymmetry metric
$\mathcal{Q}$	resonator coherence factor
$\mathcal{G}$	geometry/material coupling factor

6 Formalized Experimental Framework

Table 2: Representative protocol mapping.

Experiment	Variable isolated
Voltage sweep	$u_{\text{EM}} \sim E^2$ scaling
Resonance sweep	$\mathcal{Q}$ dependence
Geometry reversal	asymmetry factor $\mathcal{A}$
Material swap	geometry/material factor $\mathcal{G}$
Vacuum sweep	EHD/plasma rejection
Dummy load	thermal/RF rejection
Thermal lag analysis	heat-drift rejection

7 Representative Experimental Parameter Estimates

Table 3: Illustrative parameter ranges for EVRT-style residual-force metrology.

Parameter	Representative range
Voltage	10–50 kV
Frequency	MHz–GHz
Loaded resonator $\mathcal{Q}$	10^3–10^6
Pressure	atmosphere to 10^{-6} Torr
Force sensitivity	nN–μN
Candidate upper bound	$\chi_{\text{eff}} \lesssim 10^{-9}\,\text{m}^{-1}$
Thermal resolution	sub-0.1°C preferred

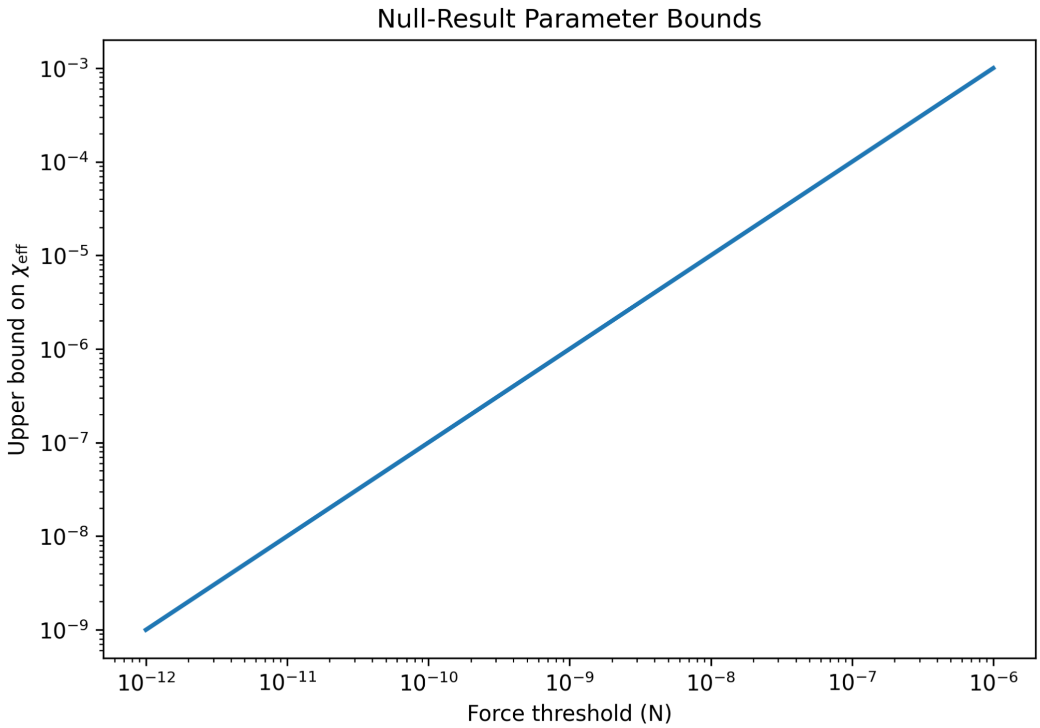

Figure 5: Illustrative null-result parameter bounds showing representative upper constraints on χ_{eff} as a function of force sensitivity threshold.

8 Inference and Parameter Extraction

The effective response coefficient may be inferred through

$$\chi_{\text{eff}} = \frac{F_{\text{res}}}{u_{\text{EM}}\, V\, \mathcal{A}\, \mathcal{Q}\, \mathcal{G}}.$$

Equation (3) defines the inferred effective response coefficient under measured residual conditions.

For null results,

$$\chi_{\text{eff}} < \frac{F_{\text{threshold}}}{u_{\text{EM}}\, V\, \mathcal{A}\, \mathcal{Q}\, \mathcal{G}}$$

Equation (4) defines the corresponding upper-bound constraint under null-result conditions.

9 Representative Diagnostic Structures

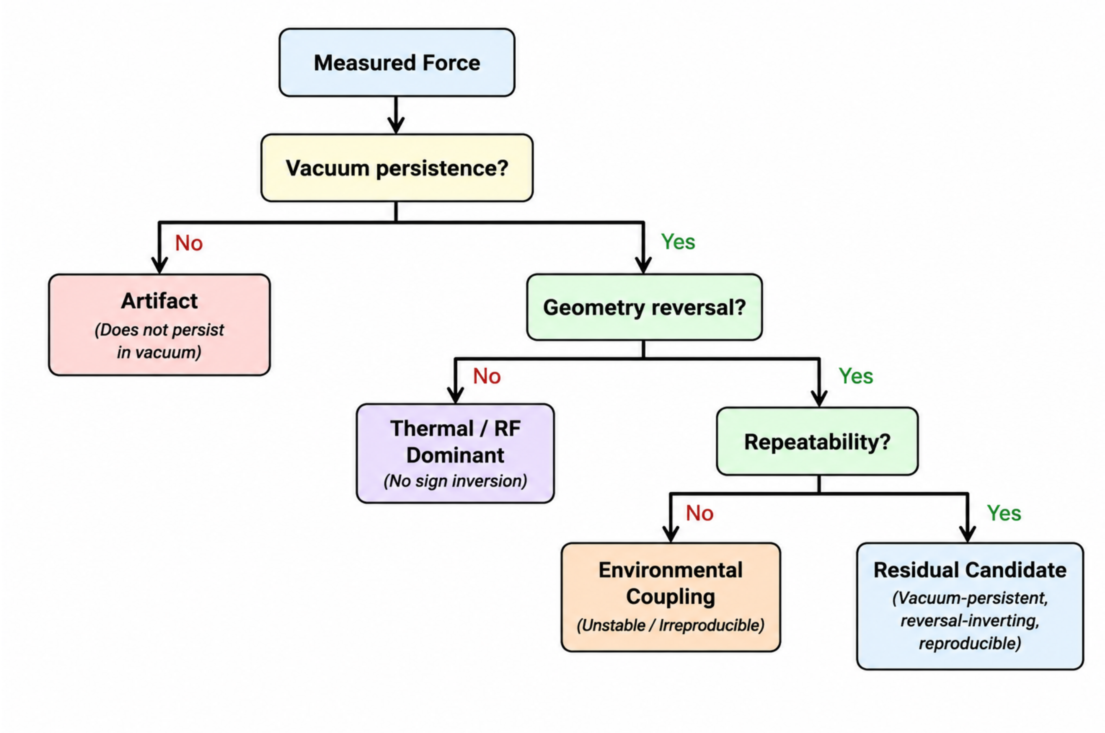

Figure 6: Illustrative diagnostic decision structure for systematic residual-force classification and artifact rejection.

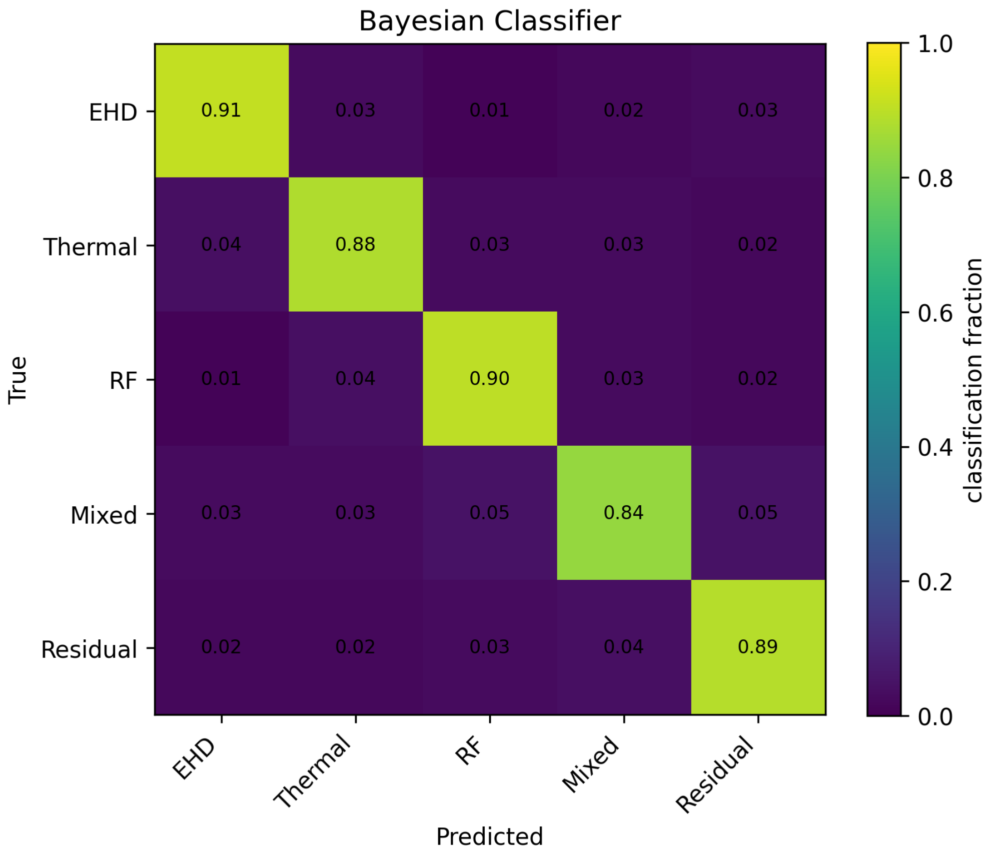

Figure 7: Illustrative Bayesian diagnostic classifier structure demonstrating conceptual separation between thermal, RF, EHD, mixed, and residual candidate signatures.

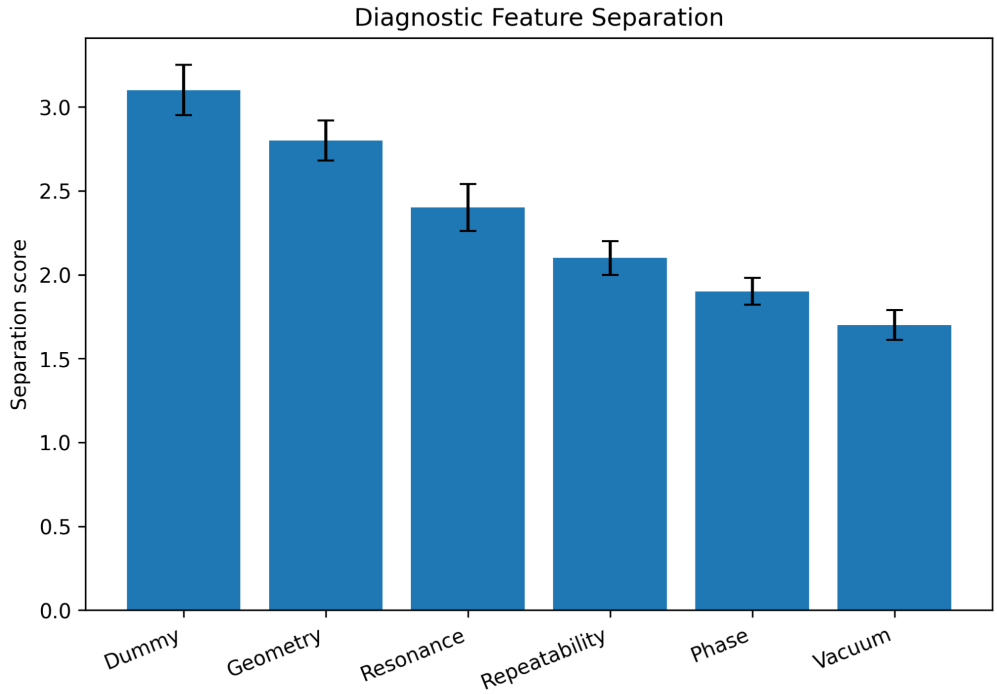

Figure 8: Representative diagnostic-feature separation metrics for candidate residual classification pathways.

10 Candidate Observable Signatures

Potential residual-candidate indicators include:

- persistence under vacuum,
- sign inversion under geometry reversal,
- potential approximate scaling with E^2,
- dependence on resonator $\mathcal{Q}$,
- weak thermal-lag correlation,
- stable repeated phase behavior.

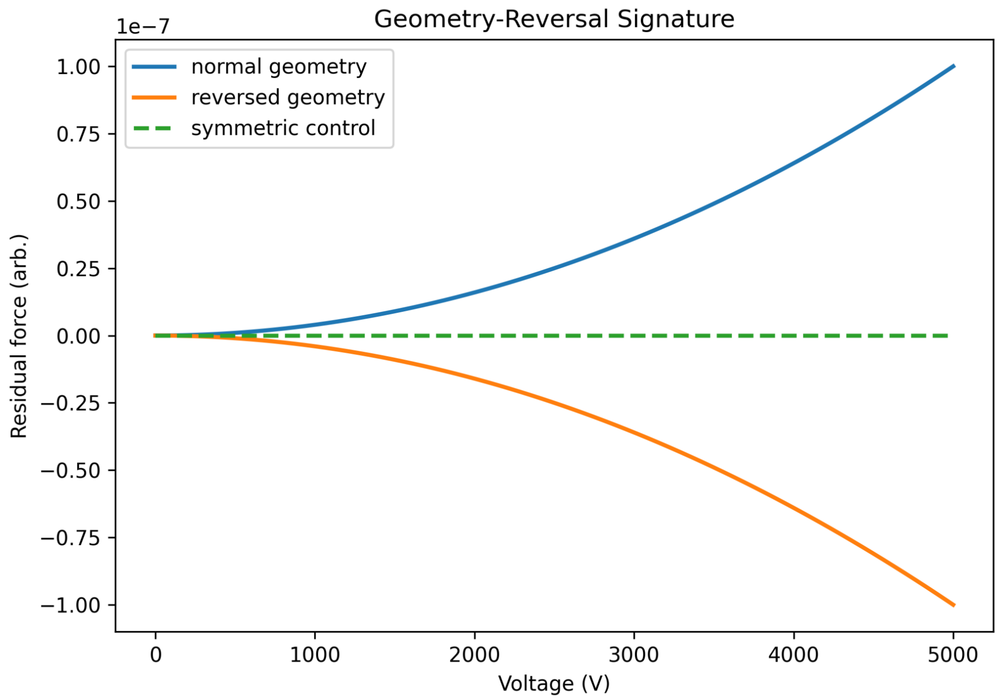

Figure 9: Illustrative geometry-reversal signature showing candidate polarity inversion under asymmetric configuration reversal.

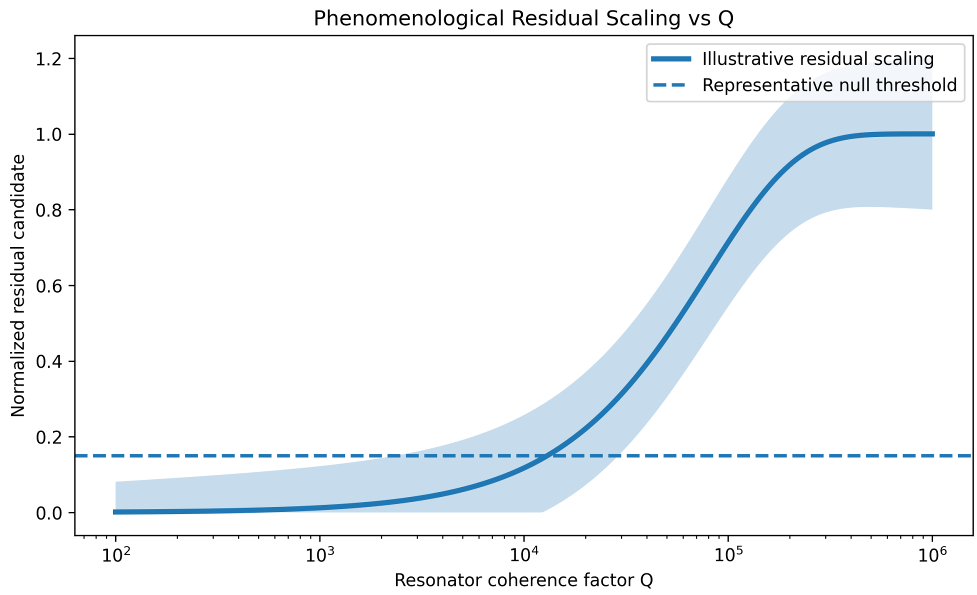

Figure 10: Illustrative phenomenological residual-force scaling versus resonator coherence factor showing representative uncertainty and saturation behavior.

11 Representative Systematic Error Sources

Important systematic uncertainty channels include:

- thermal drift,
- RF cable coupling,
- plasma and EHD effects,
- mechanical vibration,
- sensor drift,
- airflow and acoustic coupling,
- electrostatic charging effects.

12 Historical Context and Research Lineage

Historically, many claimed anomalous-force observations were later attributed to incomplete thermal, electromagnetic, or environmental isolation.

Many historical anomalous-force investigations lacked simultaneous vacuum testing, geometry reversal, dummy-load controls, thermal-lag analysis, resonance characterization, and independent replication.

The present framework attempts to combine these requirements into a unified precision-metrology structure.

13 Falsification Criteria

The EVRT residual-force hypothesis is disfavored if:

- signals collapse in vacuum,

- geometry reversal fails,
- dummy-load substitution reproduces the effect,
- thermal lag fully explains the signal,
- repeatability is not maintained.

14 Reproducibility Doctrine

Residual-force investigations are particularly vulnerable to correlated systematic effects because multiple environmental and electromagnetic channels may scale simultaneously with operating power, resonance conditions, or geometric asymmetry. Consequently, reproducibility requirements must extend beyond simple signal persistence and include multi-domain diagnostic agreement across independent experimental configurations. Any residual-force claim should require:

- independent replication,
- geometry reversal,
- vacuum persistence,
- uncertainty-budget reporting,
- multi-domain diagnostic agreement.

15 Measurement Workflow

16 Open Experimental Questions

Important unresolved questions include:

- Can geometry-reversal signatures survive high vacuum?
- Does apparent scaling persist under independent replication?
- Are residual candidates correlated with resonance-domain transitions?
- Can systematic Bayesian diagnostics reliably separate correlated artifact channels?

17 Scope of Claims

This work:

- does not claim verified propulsion,
- does not claim conservation-law violation,
- does not claim spacetime engineering,
- proposes only a phenomenological diagnostic and metrology framework.

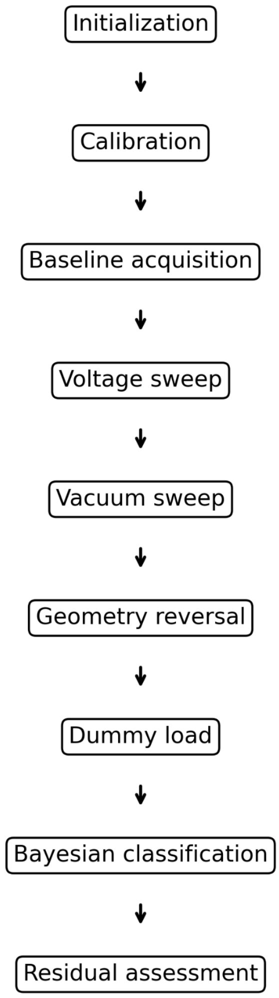

Figure 11: Illustrative example workflow for precision residual-force metrology testing.

18 Conclusion

No experimentally verified residual-force effect is claimed in this work.

The framework instead defines experimentally testable conditions under which residual-force candidates may be constrained, classified, or investigated.

Even under null results, the framework remains scientifically useful because it constrains residual-response parameter space while improving precision electromagnetic metrology methodology.

The present framework is intended primarily as a falsifiable precision-metrology architecture rather than evidence for established anomalous propulsion phenomena.

Future implementations may benefit from Bayesian multi-domain diagnostic classification to systematically separate correlated artifact channels from candidate residual signatures.

A Example Experimental Configuration

A minimal implementation could use:

- asymmetric resonant cavities,
- high-voltage asymmetric capacitor systems,
- torsion-balance readout,
- vacuum compatibility,
- synchronized multi-domain acquisition.

B Nomenclature

χ_{eff}	Effective residual-response coefficient
u_{EM}	Electromagnetic energy density
$\mathcal{A}$	Asymmetry factor
$\mathcal{Q}$	Resonance/coherence factor
$\mathcal{G}$	Geometry/material coupling factor
F_{res}	Residual-force candidate

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